

Units and Dimensions in Electromagnetism

A circuits-focused dimensional toolkit

The goal of this handout is speed: by the end you should be able to glance at any combination of R , L , C , ω , q , B , ϵ_0 , μ_0 , σ and instantly name what kind of quantity it is. The text states the essential facts; the points are won in the problems. There is a total of **66** points. Problems marked **[A]** are harder.

1 The Currency of a Circuit

Idea 1: Three units run every circuit

Every electrical quantity in a lumped circuit is built from just three SI units: the **volt** V, the **amp** A, and the **second** s. The defining relations $V = IR$, $Q = CV$, $V = L dI/dt$, and $I = dQ/dt$ fix the rest:

$$[R] = \frac{\text{V}}{\text{A}}, \quad [C] = \frac{\text{A s}}{\text{V}}, \quad [L] = \frac{\text{V s}}{\text{A}}, \quad [Q] = \text{A s}, \quad [\Phi] = \text{V s},$$

$$[\text{power}] = \text{VA}, \quad [\text{energy}] = \text{VA s}.$$

Notice the pleasing symmetry between C and L : capacitance is “amp-seconds per volt” and inductance is “volt-seconds per amp”. They are mirror images, and that mirror is the key to almost everything below.

Remark

A capacitor *hoards charge*: $C = Q/V$ tells you how many amp-seconds it banks per volt. An inductor *hoards flux*: $L = \Phi/I$ tells you how many volt-seconds it banks per amp. “Flux” $\Phi = \int V dt$ is just volt-seconds—an inductor is to current what a capacitor is to voltage.

[7] Problem 1. Express the SI unit of each quantity using only volts V, amps A, and seconds s:

- (a) resistance R
- (b) capacitance C
- (c) inductance L
- (d) charge Q
- (e) magnetic flux Φ
- (f) power P
- (g) energy U

[4] Problem 2. A reactance is the “effective resistance” of an inductor or capacitor to AC. Show on dimensional grounds that the inductive reactance ωL and the capacitive reactance $1/(\omega C)$ both come out in ohms, so it makes sense to add them to R when forming an impedance.

2 Why RC , L/R , and $1/\sqrt{LC}$ Are All Times

Idea 2: Read the time off the defining relation

You never need to memorize that RC and L/R are times—each falls out of one physical sentence.

RC . A capacitor holds charge $Q = CV$. A resistor drains it at a current $I = V/R$. The draining time is

$$\tau \sim \frac{\text{charge stored}}{\text{rate of draining}} = \frac{CV}{V/R} = RC.$$

“How much you store” over “how fast you remove it” is always a time.

L/R . An inductor carries current I ; the resistor needs a voltage IR to push against the inductor’s $V = L dI/dt$. Balancing them, $L dI/dt \sim IR$, so I changes on the timescale

$$\tau \sim \frac{L}{R}.$$

Idea 3: One formula to rule them all

Every decay constant in a passive circuit is

$$\tau = \frac{\text{energy stored}}{\text{power dissipated}}$$

which is manifestly a time (joules over watts). For the capacitor, $\tau = \frac{1}{2}CV^2/(V^2/R) = \frac{1}{2}RC$. For the inductor, $\tau = \frac{1}{2}LI^2/(I^2R) = \frac{1}{2}L/R$. The numerical factor depends on your definition; the *units* never lie.

Idea 4: $\sqrt{LC}$ is the geometric mean of the other two times

The cleanest reason $\sqrt{LC}$ is a time requires no new physics at all:

$$LC = \left(\frac{L}{R}\right)(RC) \implies \sqrt{LC} = \sqrt{\frac{L}{R} \cdot RC}.$$

The right side is the geometric mean of two times, hence a time. The resistance R cancels—it appears once on top (in L/R) and once on the bottom (in RC)—which is exactly why the LC resonant period doesn’t care about the resistor. R only damps the oscillation; it cannot set its frequency.

Example 1: Energy-balance derivation of $\sqrt{LC}$

Forget the geometric mean and rebuild $\sqrt{LC}$ from energy. In an LC oscillation energy sloshes back and forth, so at peak the capacitor’s $\frac{1}{2}Q^2/C$ equals the inductor’s $\frac{1}{2}LI^2$. Over one quarter-period τ the current I delivers a charge $Q \sim I\tau$. Substituting,

$$\frac{(I\tau)^2}{C} \sim LI^2 \implies \tau \sim \sqrt{LC}.$$

The same answer, now seen as the time for current to refill the capacitor's charge.

- [3] **Problem 3.** Using only a resistor R and capacitor C , prove that the *only* power-law combination $R^a C^b$ with units of time is RC itself (i.e. $a = b = 1$). Do the same to show L/R is the unique time built from L and R .
- [4] **Problem 4.** The geometric-mean argument says $\sqrt{LC} = \sqrt{(L/R)(RC)}$ for *any* R . Pick two wildly different resistors, $R_1 = 1\ \Omega$ and $R_2 = 10^6\ \Omega$, in a circuit with $L = 1\ \text{mH}$ and $C = 1\ \text{nF}$. Compute L/R and RC for each, verify their product is the same, and state in one sentence why this had to happen.
- [4] **Problem 5.**  The five formulas below have each been garbled. Use dimensions alone to spot the error and write the corrected form (numerical prefactors don't matter):

- (a) resonant frequency $\omega_0 = \frac{1}{LC}$
- (b) RC decay time $\tau = RC^2$
- (c) energy in a capacitor $U = \frac{1}{2}CV$
- (d) series-RLC quality factor $Q = R\sqrt{L/C}$
- (e) skin depth $\delta = \frac{1}{\mu_0\sigma\omega}$

3 $\sqrt{L/C}$ Is a Resistance

Idea 5: The other combination of L and C

If $\sqrt{LC}$ is a time, what is $\sqrt{L/C}$? Dividing instead of multiplying,

$$\frac{L}{C} = \frac{\text{Vs/A}}{\text{As/V}} = \frac{\text{V}^2}{\text{A}^2} \implies \sqrt{\frac{L}{C}} = \frac{\text{V}}{\text{A}} = \text{ohms}.$$

This is the **characteristic impedance** $Z_0 = \sqrt{L/C}$. So from L and C you can build exactly two things: a time $\sqrt{LC}$ and a resistance $\sqrt{L/C}$. At resonance the inductor's reactance $\omega_0 L$ and the capacitor's $1/(\omega_0 C)$ both equal this same Z_0 .

- [4] **Problem 6.** A coaxial cable has inductance $L = 250\ \text{nH/m}$ and capacitance $C = 100\ \text{pF/m}$ per unit length.
- (a) Show $\sqrt{L/C}$ is the same number whether you use the per-metre values or the totals for any fixed length, and compute the cable's characteristic impedance.
- (b) For a wave on the line, show that $1/\sqrt{L_\ell C_\ell}$ (subscript ℓ = "per unit length") is a *speed*, and explain why it must be, given that $\sqrt{LC}$ is a time.
- [3] **Problem 7.** Show that the impedance of free space $\sqrt{\mu_0/\epsilon_0}$ is a resistance, and that $1/\sqrt{\mu_0\epsilon_0}$ is a speed. Evaluate both ($\mu_0 = 4\pi \times 10^{-7}$, $\epsilon_0 = 8.85 \times 10^{-12}$ SI) and identify the speed.

4 The Quality Factor

Idea 6: Q is a ratio of two times, hence dimensionless

The quality factor measures how many oscillations a resonator rings out before its energy decays. Its universal definition is

$$Q = \omega_0 \times \frac{\text{energy stored}}{\text{power dissipated}} = \omega_0 \tau_{\text{decay}},$$

a (frequency) $\times$ (time), so dimensionless. For a series RLC the energy decay time is $\tau = L/R$, giving

$$Q = \omega_0 \frac{L}{R} = \frac{L/R}{\sqrt{LC}} = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{Z_0}{R}.$$

Read the middle expression out loud: Q is the *decay time divided by the oscillation time*—roughly the number of radians the circuit swings through before it dies. For a parallel RLC the dissipation flips and $Q = \omega_0 RC = R/Z_0$.

Remark

The series and parallel results are reciprocals, $Q_{\text{series}} = Z_0/R$ versus $Q_{\text{parallel}} = R/Z_0$. Either way Q compares the resistor to the one natural resistance in the problem, $Z_0 = \sqrt{L/C}$. High Q means R is far from Z_0 .

- [4] **Problem 8.** Show in three independent ways that the series-RLC quality factor is dimensionless: (a) from $Q = \omega_0 L/R$, (b) from $Q = (1/R)\sqrt{L/C}$, and (c) from $Q = \omega_0 \tau$ with τ the energy-decay time L/R . Confirm all three are the same combination.
- [5] **Problem 9.** An LC tank has $L = 2$ mH and $C = 5$ nF.
- (a) Find the resonant frequency f_0 and the characteristic impedance Z_0 .
 - (b) A resistor $R = 10 \Omega$ is placed in series. Find Q without computing any energies—just use Z_0 .
 - (c) Roughly how many oscillations does the circuit ring through before its energy falls by $1/e$?

5 Field Quantities and Material Constants

Idea 7: Fields, flux, and energy density

Push past circuit elements into the fields themselves:

$$[\Phi] = \text{Vs (weber)}, \quad [B] = \frac{\text{Vs}}{\text{m}^2} \text{ (tesla)}, \quad [E] = \frac{\text{V}}{\text{m}}, \quad [\varepsilon_0] = \frac{\text{As}}{\text{Vm}}, \quad [\mu_0] = \frac{\text{Vs}}{\text{Am}}.$$

The combinations to recognize on sight:

$$\varepsilon_0 E^2 \text{ and } \frac{B^2}{\mu_0} \text{ are energy densities } \left(\frac{\text{J}}{\text{m}^3} \right), \quad \frac{EB}{\mu_0} \text{ is the Poynting flux } \left(\frac{\text{W}}{\text{m}^2} \right).$$

And the master identity $\varepsilon_0 \mu_0 = 1/c^2$ is forced by units: their product is s^2/m^2 .

Idea 8: Material constants hide more times and lengths

Resistivity ρ (units $\Omega \text{ m}$) and conductivity $\sigma = 1/\rho$ generate their own scales:

$\varepsilon_0 \rho = \frac{\varepsilon_0}{\sigma}$ is a time (charge-relaxation time), $\mu_0 \sigma a^2$ is a time (magnetic-diffusion time, size a),

$\frac{1}{\mu_0 \sigma \omega}$ is an area $\Rightarrow \delta \sim \sqrt{\frac{2}{\mu_0 \sigma \omega}}$ is the skin depth.

- [3] **Problem 10.** Show that $\varepsilon_0 \rho$ has units of time. Estimate it for copper ($\rho = 1.7 \times 10^{-8} \Omega \text{ m}$) and comment on what such a fast time means physically for free charge dumped inside a metal.
- [4] **Problem 11.** Show that $\mu_0 \sigma a^2$ is a time, where a is a length. This is the time for eddy currents (and the trapped field) to decay in a conductor of size a . Estimate it for a copper sphere of radius $a = 10 \text{ cm}$ ($\sigma_{\text{Cu}} = 5.9 \times 10^7 \text{ S/m}$).
- [4] **Problem 12.** From the permeability μ_0 , the conductivity σ , and a driving angular frequency ω , construct—by dimensions alone—the unique length scale you can form. Confirm it is the skin depth and explain why higher frequency makes it smaller.
- [4] **Problem 13.** Verify both EM energy densities and the Poynting flux:
- (a) $\varepsilon_0 E^2$ has units of J/m^3 ,
 - (b) B^2/μ_0 has units of J/m^3 ,
 - (c) EB/μ_0 has units of W/m^2 .
- [4] **Problem 14.** Two important frequencies of charged-particle motion:
- (a) Show the cyclotron frequency qB/m is a frequency. (Use $[B] = \text{kg}/(\text{As}^2)$.)
 - (b) Show the plasma frequency $\sqrt{ne^2/(\varepsilon_0 m)}$ is a frequency, where n is a number density.
- [4] **Problem 15.** Classify each combination as a **time**, **frequency**, **resistance**, or **dimensionless** number:

$$RC, \quad \frac{L}{R}, \quad \sqrt{LC}, \quad \frac{1}{\sqrt{LC}}, \quad \sqrt{\frac{L}{C}}, \quad \omega L, \quad \frac{1}{\omega C}, \quad \omega RC, \quad \frac{R}{\omega L}, \quad \omega^2 LC, \quad \frac{\omega L}{R}, \quad \frac{R^2 C}{L}.$$

- [5] **Problem 16.** [A] Show that the fine-structure constant $\alpha = \frac{e^2}{4\pi\varepsilon_0 \hbar c}$ is dimensionless. (Hint: show e^2/ε_0 has units of energy $\times$ length, and so does $\hbar c$.) This is why $\alpha \approx 1/137$ is the same pure number in every unit system.

6 Solutions

Throughout, V, A, s, m denote volt, amp, second, metre, and $[X]$ means “the units of X .”

1. Straight from the defining relations. $[R] = \text{V/A}$; $[C] = [Q]/[V] = \text{As/V}$; $[L] = [V]/[dI/dt] = \text{Vs/A}$; $[Q] = \text{As}$; $[\Phi] = [LI] = \text{Vs}$ (equivalently $\int V dt$); $[P] = [VI] = \text{VA}$; $[U] = [Pt] = \text{VAs}$.
2. $[\omega L] = \frac{1}{s} \cdot \frac{\text{Vs}}{\text{A}} = \frac{\text{V}}{\text{A}} = \Omega$, and $\left[\frac{1}{\omega C}\right] = \frac{1}{(1/s)(\text{As/V})} = \frac{\text{V}}{\text{A}} = \Omega$. Both are ohms, so R , ωL , and $1/(\omega C)$ live on the same footing and can be combined (in quadrature) into an impedance.
3. Write $R^a C^b = (\text{V/A})^a (\text{As/V})^b = \text{V}^{a-b} \text{A}^{b-a} \text{s}^b$. For this to be a pure time we need the volt exponent $a - b = 0$, the amp exponent $b - a = 0$, and the second exponent $b = 1$. Hence $b = 1$ and $a = 1$: only RC . Similarly $L^a R^b = (\text{Vs/A})^a (\text{V/A})^b = \text{V}^{a+b} \text{A}^{-a-b} \text{s}^a$; a time needs $a + b = 0$ and $a = 1$, so $b = -1$: only L/R .
4. For $R_1 = 1\Omega$: $L/R = 10^{-3}/1 = 10^{-3}\text{s}$ and $RC = 1 \cdot 10^{-9} = 10^{-9}\text{s}$; product $= 10^{-12}\text{s}^2$. For $R_2 = 10^6\Omega$: $L/R = 10^{-3}/10^6 = 10^{-9}\text{s}$ and $RC = 10^6 \cdot 10^{-9} = 10^{-3}\text{s}$; product $= 10^{-12}\text{s}^2$. Same product because $(L/R)(RC) = LC$, which has no R in it. (And indeed $\sqrt{LC} = 10^{-6}\text{s}$.) Changing R just trades one time for the other while keeping their product—the resonant period—fixed.
5. (a) $1/(LC)$ has units $1/\text{s}^2$, a frequency *squared*; the resonant frequency is $\omega_0 = 1/\sqrt{LC}$. (b) RC^2 is $(\text{s})(\text{As/V}) = \text{time} \times \text{capacitance}$, not a time; should be RC . (c) CV is $(\text{As/V})(\text{V}) = \text{As} = \text{charge}$, not energy; should be $\frac{1}{2}CV^2$. (d) $R\sqrt{L/C} = \Omega \cdot \Omega = \Omega^2$, not dimensionless; the series-RLC factor is $Q = (1/R)\sqrt{L/C}$. (e) $1/(\mu_0\sigma\omega)$ is an *area* (see Problem 13), not a length; the skin depth is $\delta \sim \sqrt{2/(\mu_0\sigma\omega)}$.
6. (a) $\sqrt{L/C}$ taken from per-metre values is $\sqrt{(L_\ell x)/(C_\ell x)}$ for any length x ; the x cancels, so the impedance is a property of the cable, not its length. Numerically $Z_0 = \sqrt{250 \times 10^{-9}/100 \times 10^{-12}} = \sqrt{2500} = 50\Omega$. (b) $[1/\sqrt{L_\ell C_\ell}]$: per-length L_ℓ and C_ℓ carry an extra $1/\text{m}$ each, so $L_\ell C_\ell = (LC)/\text{m}^2$, i.e. $\text{time}^2/\text{length}^2$, and its inverse square root is m/s , a speed. It must be a speed because $\sqrt{LC}$ is a time and the only way to get a speed from a time is to divide by length—which is exactly what “per unit length” supplies.
7. $\left[\sqrt{\mu_0/\epsilon_0}\right] = \sqrt{\frac{\text{Vs/(Am)}}{\text{As/(Vm)}}} = \sqrt{\text{V}^2/\text{A}^2} = \text{V/A} = \Omega$, giving $\sqrt{4\pi \times 10^{-7}/8.85 \times 10^{-12}} \approx 377\Omega$. $[1/\sqrt{\mu_0\epsilon_0}]$: the product $\mu_0\epsilon_0$ has units s^2/m^2 , so its inverse square root is m/s , and numerically $1/\sqrt{4\pi \times 10^{-7} \cdot 8.85 \times 10^{-12}} \approx 3.0 \times 10^8 \text{m/s} = c$, the speed of light.
8. (a) $\omega_0 L/R = (1/\text{s})(\text{Vs/A})/(\text{V/A}) = 1$. (b) $(1/R)\sqrt{L/C} = (\text{A/V})(\text{V/A}) = 1$. (c) $\omega_0\tau = (1/\text{s})(\text{s}) = 1$. All equal because $\omega_0(L/R) = \frac{1}{\sqrt{LC}} \frac{L}{R} = \frac{1}{R} \sqrt{L/C}$ —the same combination written three ways.
9. (a) $\omega_0 = 1/\sqrt{LC} = 1/\sqrt{(2 \times 10^{-3})(5 \times 10^{-9})} = 1/\sqrt{10^{-11}} \approx 3.16 \times 10^5 \text{rad/s}$, so $f_0 = \omega_0/2\pi \approx 50.3 \text{kHz}$. And $Z_0 = \sqrt{L/C} = \sqrt{(2 \times 10^{-3})/(5 \times 10^{-9})} = \sqrt{4 \times 10^5} \approx 632\Omega$. (b) $Q = Z_0/R = 632/10 \approx 63$. (c) The energy falls by $1/e$ in $\tau = L/R$, and $Q = \omega_0\tau$ is the number of radians in that time, so the number of full cycles is $Q/2\pi \approx 63/6.28 \approx 10$.

10. $[\varepsilon_0 \rho] = \frac{\text{As}}{\text{Vm}} \cdot \frac{\text{Vm}}{\text{A}} = \text{s}$. For copper, $\tau = \varepsilon_0 \rho = (8.85 \times 10^{-12})(1.7 \times 10^{-8}) \approx 1.5 \times 10^{-19} \text{ s}$. Such an astonishingly short time means that any free charge placed inside a good conductor neutralizes essentially instantly—the charge flees to the surface, leaving the interior neutral, which is why electrostatics treats conductors as having zero interior field.
11. $[\mu_0 \sigma a^2] = \frac{\text{Vs}}{\text{Am}} \cdot \frac{\text{A}}{\text{Vm}} \cdot \text{m}^2 = \text{s}$. For the copper sphere, $\tau = \mu_0 \sigma a^2 = (4\pi \times 10^{-7})(5.9 \times 10^7)(0.1)^2 \approx 0.74 \text{ s}$. (Geometry adds an $O(1)$ factor.) Currents induced in the metal coast for roughly this long before ohmic losses kill them.
12. We want a length from $\mu_0 \sigma$ (units $\frac{\text{Vs}}{\text{Am}} \cdot \frac{\text{A}}{\text{Vm}} = \text{s/m}^2$) and ω (units $1/\text{s}$). The product $\mu_0 \sigma \omega$ has units $1/\text{m}^2$, so its reciprocal is an area and $\delta \sim 1/\sqrt{\mu_0 \sigma \omega}$ is the unique length—the skin depth (the standard prefactor is $\sqrt{2}$). Higher ω puts a larger number in the denominator, shrinking δ : fast-changing fields are expelled to a thinner surface layer.
13. (a) $[\varepsilon_0 E^2] = \frac{\text{As}}{\text{Vm}} \left(\frac{\text{V}}{\text{m}}\right)^2 = \frac{\text{AsV}}{\text{m}^3} = \frac{\text{VAs}}{\text{m}^3} = \frac{\text{J}}{\text{m}^3}$. (b) $[B^2/\mu_0] = \frac{(\text{Vs/m}^2)^2}{\text{Vs/(Am)}} = \frac{\text{VAs}}{\text{m}^3} = \frac{\text{J}}{\text{m}^3}$. (c) $[EB/\mu_0] = \frac{(\text{V/m})(\text{Vs/m}^2)}{\text{Vs/(Am)}} = \frac{\text{VA}}{\text{m}^2} = \frac{\text{W}}{\text{m}^2}$.
14. (a) With $[B] = \text{kg}/(\text{As}^2)$, $[qB/m] = (\text{As}) \frac{\text{kg}/(\text{As}^2)}{\text{kg}} = \frac{1}{\text{s}}$, a frequency. (b) $\left[\frac{ne^2}{\varepsilon_0 m}\right] = \frac{1}{\text{m}^3} \cdot \frac{(\text{As})^2}{[\varepsilon_0] \text{kg}} = \frac{1}{\text{m}^3} \cdot \frac{\text{A}^2 \text{s}^2 \text{Vm}}{\text{As kg}} = \frac{\text{AsV}}{\text{m}^2 \text{kg}}$. Since $\text{AV} = \text{W} = \text{kg m}^2/\text{s}^3$, this is $\frac{\text{s} \cdot \text{kg m}^2/\text{s}^3}{\text{m}^2 \text{kg}} = \frac{1}{\text{s}^2}$, so the square root is a frequency.
15. $RC = \text{s}$ (time); $L/R = \text{s}$ (time); $\sqrt{LC} = \text{s}$ (time); $1/\sqrt{LC} = 1/\text{s}$ (frequency); $\sqrt{L/C} = \Omega$ (resistance); $\omega L = \Omega$ (resistance); $1/(\omega C) = \Omega$ (resistance); ωRC : $(1/\text{s})(\text{s}) = \text{dimensionless}$; $R/(\omega L)$: $\Omega/\Omega = \text{dimensionless}$; $\omega^2 LC$: $(1/\text{s}^2)(\text{s}^2) = \text{dimensionless}$; $\omega L/R$: $\Omega/\Omega = \text{dimensionless}$ ($= Q$); $R^2 C/L$: $\Omega^2 \cdot (\text{F/H}) = \Omega^2/\Omega^2 = \text{dimensionless}$ ($= 1/Q^2$).
16. $[e^2/\varepsilon_0] = \frac{(\text{As})^2}{\text{As}/(\text{Vm})} = \text{As} \cdot \text{Vm} = (\text{VAs}) \text{m} = \text{J} \cdot \text{m}$, energy $\times$ length. Likewise $\hbar c$ has units $(\text{Js})(\text{m/s}) = \text{J} \cdot \text{m}$. Their ratio $\frac{e^2}{4\pi\varepsilon_0 \hbar c}$ is therefore a pure number, the same $\approx 1/137$ in any system of units.